

CIG EMPP: AI-Powered Media Vault Platform

A Scalable, Cloud-Native Architecture for Campus Media
Management

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| Enterprise-Grade Tech Stack



Frontend : Next.js, Tailwind CSS, Lucide Icons. Responsive Glassmorphism UI.



Backend : Node.js, Express, JWT. Strict separation of concerns API.



Database : Neon PostgreSQL & Prisma ORM. 3NF Normalized Schema.



Machine Learning : AWS Rekognition Engine (Computer Vision & Biometric Vector Matching).



Processing : Sharp (High-speed, C++ based Node.js image compiler)



Infrastructure : Full CI/CD via Vercel and Render cloud ecosystems.

| Architecture Core 1 - Ingestion

The Server Bottleneck: Traditional file uploads buffer in server RAM (OOM crashes) or local disks, throttling concurrent users.

The Presigned URL Solution:

- Client requests a cryptographic, time-limited (60s) upload token via `getSignedUrl`.
- Node.js authorizes the JWT and generates the AWS token.
- Client pipes the raw binary data directly into the AWS S3 Bucket via a PUT request.
- Impact: Reduces backend bandwidth consumption by 99%. Enables simultaneous 4K video uploads from hundreds of users.

| Architecture Core 2 - Machine Learning

Trigger Mechanism: Asynchronous background tasks triggered post-S3-ingestion.

Smart Tagging (`DetectLabelsCommand`): Employs deep learning to identify scenes, objects, and concepts with a >75% Confidence Threshold (e.g., #Photography, #Crowd).

Facial Recognition (`SearchFacesByImage`):

- Extracts 128-dimensional facial vectors from event photos.
- Computes cosine similarity against the secure cig-faces-collection.
- Automatically links the Postgres `UserId` to the Media record if a match exceeds 85% confidence.

| Architecture Core 3 - Secure Distribution

S3 Bucket Obfuscation: The raw S3 bucket blocks all public access to prevent unauthorized scraping.

Secure Stream Proxy: Media is piped directly through the Node.js API (`s3Response.Body.pipe(res)`) ensuring memory-efficient delivery.

On-The-Fly Watermarking:

- Intercepts download requests.
- Reads image dimensions dynamically via image metadata.
- Generates a responsive SVG buffer containing: [Club Name | Event Name | User Role].
- Composites layers using sharp before serving the attachment.

| Database & Data Integrity

- **Schema Design:** Fully normalized PostgreSQL database designed for complex 1-to-Many relationships.
- **Hierarchical Structure:** Event -> Album -> Media.
- **Data Integrity:** Implemented cascading cleanup scripts to prevent orphaned data (e.g., deleting a media file automatically purges its associated comments and likes).
- **Array Handling:** Native Postgres array columns utilized for tags and facesInPhoto for ultra-fast client-side filtering.

| AI/ML Workflow: Rekognition Engine

Phase 1: Smart Tagging

Automatic contextual labeling
(#Crowd, #Sports).

Phase 2: Face Enrollment

Biometric vector extraction from
user selfies.

Phase 3: Automated Discovery

Real-time matching against the
AWS Face Collection.

Perks

Cloud-Native: Offload storage to AWS S3
and computation to AWS AI.

Zero-Bottleneck: Prevent crashes via direct-
to-cloud streams.

Automation: Replace manual tagging with
Machine Learning.

| Security & Dynamic Processing

RBAC & Privacy

- Role-Based Access Control (Admin to Viewer).
- Private media hidden via **Backend S3 Proxy**.
- Raw S3 links cryptographically secured.

Dynamic Watermarking

On-the-fly processing using **sharp**.
Server fetches S3 buffer and overlays a dynamic SVG watermark based on Role and Metadata in milliseconds.

Social Engagement & UI/UX

- Global Search: Instant filters by Event, Tag, Date.
- Social Feed: Hearts, Likes, and manual tagging.
- Real-time Notifications: Alert system for AI Face Recognition matches.
- Sandbox Mode: Switch roles for testing interactions.

Conclusion & Production Readiness

- ✓ Fully Deployed: Operational on Vercel, Render, and Neon.
- ✓ Problem Solved: 100% of the platform requirements met.

Future Scope

- Socket.io for live WebSocket notification streams.
- Infinite scrolling pagination for massive gallery datasets.